

1. Administrative & Study Identification

Registry Study Title: Belzutifan/MK-6482 for the Treatment of Advanced Pheochromocytoma/Paraganglioma (PPGL), Pancreatic Neuroendocrine Tumor (pNET), Von Hippel-Lindau (VHL) Disease-Associated Tumors, Advanced Gastrointestinal Stromal Tumor (wt GIST), or Solid Tumors With HIF-2α Related Genetic Alterations (MK-6482-015) **Full Study Title:** A Phase 2 Study to Evaluate the Efficacy and Safety of Belzutifan (MK-6482, Formerly PT2977) Monotherapy in Participants With Advanced Pheochromocytoma/Paraganglioma (PPGL), Pancreatic Neuroendocrine Tumor (pNET), Von Hippel-Lindau (VHL) Disease-Associated Tumors, Advanced Gastrointestinal Stromal Tumor (wt GIST), or Advanced Solid Tumors With HIF-2α Related Genetic Alterations **Short Title/Acronym:** LITESPARK-015 (MK-6482-015) **Protocol Number:** MK-6482-015 / 6482-015 **NCT Number:** NCT04924075 **Posting ID:** 523016004026515442 **Sponsor / Company:** Merck (Merck Sharp & Dohme LLC) **Investigational Product:** belzutifan (MK-6482) **Trade Name:** Welireg **Phase of Development:** PHASE2 **Trial Status:** RECRUITING **Medical Monitor:** Clinical Director, Merck Sharp & Dohme LLC

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the objective response rate (ORR) of belzutifan per Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST 1.1) as assessed by Blinded Independent Central Review (BICR). **Secondary Objectives:** To evaluate the duration of response (DOR), time to response (TTR), disease control rate (DCR), progression-free survival (PFS), overall survival (OS), safety, and reduction in concomitant antihypertensive medications for applicable cohorts.

2.2 Background & Rationale To address the high unmet medical need for patients with advanced PPGL, pNET, VHL disease-associated tumors, wt GIST, or solid tumors with HIF-2α related genetic alterations. Von Hippel-Lindau (VHL) gene inactivation leads to the harmful activation of hypoxia-inducible factors (HIFs). Belzutifan is an oral HIF-2α inhibitor that targets this specific underlying mechanism, offering a targeted therapeutic option to halt tumor growth or shrink tumors in these rare oncologic diseases.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm, multi-cohort **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Targeted Enrollment: 322 participants **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 12-Aug-2021 **Estimated Primary Completion:** 04-Jun-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 12 years (adults and eligible pediatric participants depending on the cohort). 2. Documented histopathological diagnosis of advanced PPGL, pNET, VHL disease-associated tumors, wt GIST, or solid tumors with HIF-2 α related genetic alterations. 3. Measurable disease verified by blinded independent central review (BICR) per RECIST v1.1.

4.2 Exclusion Criteria 1. Unable to swallow orally administered medication or has a disorder that might affect the absorption of belzutifan. 2. History of a second malignancy, unless potentially curative treatment has been completed with no evidence of malignancy for 2 years. 3. Clinically significant cardiac disease (e.g., NYHA Class III or IV) or requiring chronic supplemental oxygen (pulse oximeter reading $<92\%$ at rest).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Drug Name: belzutifan (Trade Name: Welireg) **ATC Code:** L01XX74 **Dosage/Frequency:** 120 mg once daily (QD) for adults and pediatric patients weighing ≥ 40 kg; 80 mg once daily (QD) for pediatric patients < 40 kg. **Route of Administration:** Oral **Duration of Treatment:** Administered continuously until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Stable antihypertensive medications (must be adequately controlled with no change in medication for at least 2 weeks prior to study initiation). **Prohibited:** Prior treatment with chemotherapy, targeted therapy, biologics, or other investigational therapies within 4 weeks of the first dose of the study intervention (except somatostatin analogs).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Baseline Tumor Imaging, Safety Labs Baseline
Day 0	Registration, First Dose of Belzutifan, Vital Signs		
Interim	Every 3-4 Weeks	Safety Labs, Efficacy Assessment (Tumor Imaging every 8-12 weeks), AE recording	End of Study
At Progression	Final Vital Signs, Exit Interview, IP Accountability, Survival Follow-up		

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Objective Response Rate (ORR), defined as the percentage of participants whose tumors either completely disappear or show a meaningful reduction in size. **Measurement Method:** RECIST v1.1 criteria as assessed by Blinded Independent Central Review (BICR).

7.2 Secondary & Exploratory Endpoints 1. Duration of Response (DOR) 2. Progression-Free Survival (PFS) and Overall Survival (OS) 3. Time to Response (TTR) and Disease Control Rate (DCR)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard continuous clinical assessment and laboratory monitoring (tracking common AEs such as anemia, fatigue, musculoskeletal pain, and transaminase elevations). **Serious Adverse Events (SAEs):** Standard reporting timelines to Sponsor (typically within 24 hours of site awareness). **Data Monitoring Committee (DMC):** External and internal safety monitoring per Sponsor protocols.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy endpoints; Safety Set for all patients who received at least one dose of the study drug. **Sample Size Justification:** Target N = 322 appropriately powered to detect meaningful objective response rates across the multi-cohort design. **Handling of Missing Data:** Time-to-event endpoints (e.g., PFS, DOR, OS) will be evaluated using the Kaplan-Meier method with censoring rules applied at the last evaluable assessment.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure data integrity and protocol adherence. **Audit Trail:** eCRF locking, tracking of protocol deviations, and data clarification forms via standard quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: MK-6482-015 **Registry Number:** NCT04924075 **Sponsor/Lead Organization:** Merck (Merck Sharp & Dohme LLC) **Study Title:** LITESPARK-015 **Source Level:** 5 **Official Regulatory Title:** A Phase 2 Study to Evaluate the Efficacy and Safety of Belzutifan

(MK-6482, Formerly PT2977) Monotherapy in Participants With Advanced Pheochromocytoma/Paraganglioma (PPGL), Pancreatic Neuroendocrine Tumor (pNET), Von Hippel-Lindau (VHL) Disease-Associated Tumors, Advanced Gastrointestinal Stromal Tumor (wt GIST), or Advanced Solid Tumors With HIF-2 α Related Genetic Alterations

12. Clinical Framework (Study Specific)

Disease Areas: Pheochromocytoma/Paraganglioma, Pancreatic Neuroendocrine Tumor, Von Hippel-Lindau Disease, Advanced Gastrointestinal Stromal Tumor, HIF-2 α Mutated Cancers **Primary Condition:** Advanced PPGL, pNET, VHL Disease-Associated Tumors, wt GIST, or HIF-2 α mutated solid tumors **Diagnosis Threshold:** Histopathologically confirmed disease with measurable lesions per RECIST 1.1 **Medical Methodology:** Targeted therapy via HIF-2 α inhibition **Optimization Target:** Objective Response Rate (ORR) and Tumor Shrinkage

12.1 Disease Ontology & Semantic Classifications Included *conditions and potential disease extensions such as central nervous system tumors related to specified pathways:* * **Name / Preferred Name:** Glioma * **Preferred UMLS Name:** Genetic Neoplasms * **Semantic Type:** Neoplastic Process * **Definition:** The presence of a glioma, which is a neoplasm of the central nervous system originating from a glial cell (astrocytes or oligodendrocytes). * **MeSH Code:** D005910 * **HPO Codes:** HP:0009733 * **SNOMED ID:** 393564001 * **SNOMED Hierarchy:** 55342001|Neoplastic disease|Neoplastic disease (disorder), 64572001|Disease|Disease (disorder)

13. Study Procedures & Methodology

Management Pathway: Daily oral monotherapy combined with scheduled radiological imaging **Education Protocol (HCP):** RECIST 1.1 training, AE management (particularly anemia and hypoxia) **Education Protocol (Patient):** Oral drug administration compliance and safety reporting **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) of imaging scans to objectively confirm tumor responses

14. Observation & Follow-Up Schedule

Total Duration: Estimated 94 months (August 2021 – June 2029) **Onsite Visit Cadence:** Standard oncology cycle visits (e.g., Every 3 to 4 Weeks) **Remote Monitoring Frequency:** As required for safety and survival follow-up **Checklist Review Interval:** Radiological tumor imaging every 8 to 12 weeks

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					
Statistician	[Name]	_____	[DD-MMM-YYYY]					